

Group-based Object Alignment in Virtual Reality Environments

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ABSTRACT

Group-based object alignment is an essential manipulation task, particularly for complex scenes. In conventional 2D user interfaces, such alignment tasks are generally achieved via a command/menu-based interface. Virtual reality (VR) head-mounted displays (HMDs) provide a rich immersive interaction experience, which opens more design options for group-based object alignment interaction techniques. However, object alignment techniques in immersive environments are underexplored. In this paper, we present four interaction techniques for 3 degrees-of-freedom translational alignments: *AlignPanel*, *AlignWidget*, *AlignPin*, and *AlignGesture*. We evaluated their performance, workload, and usability in a user study with 20 participants. Our results indicate different benefits and drawbacks of these techniques for group-based alignment in immersive systems. Based on the findings, we distill a set of design choices and recommendations for these techniques in various application scenarios.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Empirical studies in interaction design**; **Interaction techniques**.

KEYWORDS

virtual reality, manipulation task, group object alignment, interaction techniques, user studies

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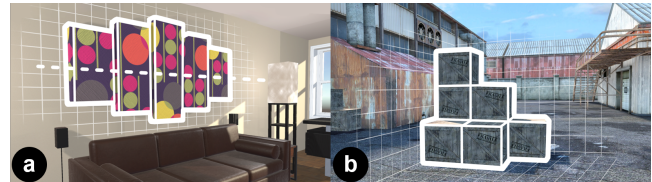


Figure 1: Two example group-based alignment scenarios. They are positioned based on the intrinsic alignments of the object group: (a) middle and back of the decorations, and (b) left, back, and bottom of the boxes.

1 INTRODUCTION

Object alignment is an essential manipulation task in content creation applications. Alignment tasks have the goal of changing one or more objects' positions according to at least one spatial constraint. When aligning multiple objects, such spatial constraints can be either: (1) derived from an object, for example, to align the right-side surface of an object or a group of objects to the left-side surface of another object; or (2) derived from the group itself, for example, to align two objects to their rightmost surfaces, or to align all group members to their geometric center along a specific axis. This difference yields different interaction processes, i.e., introduces different tasks. For the first case, users generally need to specify the (extrinsic) constraint first and then apply that constraint to one or more objects. For the second case, which we call *group-based alignment*, users need to select a group of objects first and then apply the alignment constraint to the group, as the constraint is implicitly formed by having all the objects in this group match the same final (intrinsic) alignment goal (see Fig. 1).

In 2D user interfaces (UIs), group-based alignment is generally achieved via a command/menu-based interface, which provides several align options [29, 43]. Yet, the common 2D UI interaction metaphors, i.e., mouse-click for a desktop setup and finger/stylus touch for a smartphone/tablet setup, limit the design space of group-based alignment. In contrast, immersive virtual reality (VR) environments afford interaction metaphors based on 6 degrees of freedom (6-DOF, including 3-DOF translational and 3-DOF rotational movements), which are more similar to the way we interact with objects in the real world [23, 30, 38]. VR thus allows for (more) 'direct manipulation', which yields additional design possibilities for group-based object alignments. However, current VR modeling or content creation applications only support the alignment of a

single object to another. To the best of our knowledge, there is little existing work on object alignment techniques in immersive VR environments, and the interaction techniques to support group-based alignments are underexplored. We aim to fill this research gap in this paper and, as a first step, we focus on 3-DOF translational alignments.

Towards this goal, we first specify the 3-DOF translational group-based alignment problem space and derive three criteria that the design of the interaction techniques for alignment tasks should meet. Based on these design criteria, we then introduce four new techniques—*AlignPanel*, *AlignWidget*, *AlignPin*, and *AlignGestures* for VR head-mounted displays (HMDs). Then, we conduct a user study with 20 participants to compare the user performance, workload, and experience with the four proposed techniques. Finally, we present a discussion of the results and point to design recommendations and directions for future work before concluding the paper.

In short, the contributions of this paper include: (1) The design of four interaction techniques for 3-DOF group-based object alignment tasks in VR HMDs and other immersive systems. (2) An empirical study with twenty participants, pointing to design choices and recommendations for the practical use of the proposed techniques.

2 RELATED WORK

2.1 Object Manipulation in VR

Virtual Hand is one of the primary input paradigms in immersive systems, e.g., with VR HMDs [18, 23]. Common input devices such as hand-held controllers and optical tracking modules afford 6-DOF tracking, comprising 3 translational axes and 3 rotational axes, and provide direct and natural interactions using virtual hand in a VE similar to the interactions in the real world [30]. Researchers have developed enhanced or extended versions of the virtual hand technique to make it more functional in different contexts (e.g., *Go-Go* [27], *PRISM* [8], and *HOMER* [40]). Nowadays, the virtual hand approach or its extensions are commonly used for object manipulation tasks in immersive VR environments.

LaViola et al. [18] defined selection, positioning, rotation, and scaling as the four canonical manipulation tasks for 3D UIs. Within VR contexts, translation and rotation are the two most frequent manipulation tasks investigated in the literature [2]. There have been lots of discussions on whether to integrate or separate the DOF controls for object translation and rotation tasks. On the one side, researchers proposed to control the translational and rotational DOF at the same time because integrated manipulation control has a parallel, interdependent structure and retains the “naturalness” and “intuitiveness” of people’s behaviors [12, 23, 27, 38]. On the other side, results from empirical studies have shown that 6-DOF manipulation, though suitable for fast coarse transformation, cannot offer precise manipulation [24]. Separating the translational and rotational controls can improve accuracy—for example, completing the task in a single dimension (i.e., 1-DOF) individually can ensure higher levels of precision [24, 25, 37]. Given the benefits of separating DOF, researchers have considered it as a feature when designing mid-air interaction techniques (e.g., *DS3* [22], *MAiOR* [25], and *Plane, Ray, and Point* [14]).

In this paper, we explore mid-air interactions for aligning virtual objects in an immersive VR system. An object alignment task, by its nature, typically has an implicit demand for precision. Thus, we considered DOF separation and, as a first step to fill the research gap, we focus on 3-DOF translational alignments. In addition, when multiple alignment requirements exist, we decompose the 3-DOF task into several 1-DOF sub-tasks to help users achieve the highest possible precision.

2.2 2D and 3D Alignment Tools

Alignment or other positioning manipulation techniques have been well studied in 2D UIs. The alignment approaches in these tools can be broadly categorized into snapping-based, command-based, and constraint-based. Command/menu-based techniques are most common in current commercial applications [29]. In the 1990s, *Briar* [11] combined the *snap-dragging* technique [4] and constraints to support 2D drawings, where the snapping function would assist the user in making quick alignments according to specified constraints. The *Beyond Snapping* approach [5] helped users make persistent and tweakable alignments with user-defined *StickLines* in graphics editing applications. *GACA* [43], on the other hand, used commands to achieve group-aware arrangements of graphical elements. In multi-touch displays, Frisch et al. [9] implemented *Grids & Guides* to support continuous and flexible alignments by snapping the objects to interactive grids and multi-touch alignment guides. More recently, they proposed *NEAT* to support alignment and other layout tasks using bi-manual gestures [10]. *Rock & Rails* [39] is another gesture-based manipulation system. It facilitates precise graphical layouts with a set of hand-shape gestures in combination with direct manipulation of objects.

Unlike the broad exploration of such interaction techniques in 2D environments, little prior work has been conducted on the design and development of alignment tools for 3D environments, especially for immersive environments. Song et al. [33] designed a *handle bar* metaphor to support object manipulation for desktop VR using freehand gestures. With a *handle bar*, users can manipulate and align multiple objects along a straight line (i.e., a bar). The *Context-based 3D grids* [1] enable users to perform precise mid-air manipulations in AR HMDs by displaying a reference grid relative to the hand. Hayatpur et al. [14] proposed *Plane, Ray, and Point*, which utilized symbolic gestures to specify (temporary) constraints for precise spatial manipulations. It enables quick alignment of virtual objects against plane constraints.

2.3 Multiple Object Manipulation

Many interaction techniques have been proposed for manipulating a single object at a time. While repeating the same manipulation for multiple objects is possible, it would also be more time-consuming and tedious than grouping the objects first and manipulating them together, especially when there are many objects to be aligned [21]. *Dual constraints* [35] afforded quick creation of object groups and direct manipulation of the groups in desktop VE. By dual constraining the objects nearby, the objects in the group can then be moved, rotated, and regrouped together [32, 35]. When working with virtual objects that have some association with one another, the hierarchical relationship of the group implicitly defined the

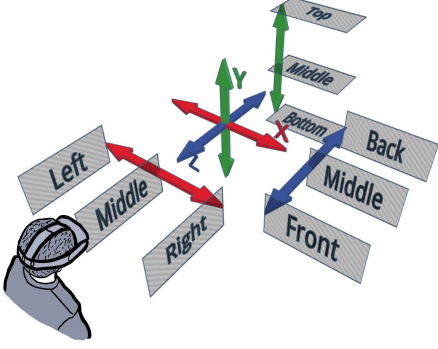


Figure 2: All nine potential alignment specifications: to align left, middle, and right along the x-axis; to align top, middle, and bottom along the y-axis; and to align front, middle, and back along the z-axis.

constraints, and thus users can interact with the object group or its sub-group(s) (e.g., [42]).

In the *Beyond Snapping* approach, users can snap the target graphical objects on pre-defined *StickyLines* and then apply alignment or distribution relationships to them [5]. Similarly, *handle bar* attaches virtual objects and manipulate them together [33]. In these two metaphors, the grouping and the group-based manipulations are achieved via explicitly formed line constraint(s). With *Plane*, *Ray*, and *Point*, users can create a plane constraint to align the objects in immersive VR environments [14]. When using these techniques, users need to first specify the constraint(s) explicitly. The constraint(s) are then used for forming a group and performing follow-up manipulations. In our work, we focus on a different problem space, where users are aiming to align multiple objects based on their implicitly formed constraints.

3 PROBLEM FORMULATION

In this paper, we focus on 3-DOF translational group-based alignments in VR environments. For simplicity, we investigate spatial alignments along the 3D cardinal directions, as shown in Fig. 2. Rotating the coordinate system then naturally affords alignment to other axes, which is a general strategy used in commercial applications (e.g., [16]). In our work, we define the term ‘group-based alignment’ as adjusting the positions of the selected objects precisely according to the furthest surface in a specific direction (align left, right, top, bottom, front, or back) or their geometric centers (align middle and along x-, y-, or z-axis). The alignment is discrete and sequential, which means that each alignment is an individual 1-DOF operation, and the target position in an alignment is always calculated based on the current position of the object group.

In addition, when applying an alignment constraint to the target objects, the objects can be *intersecting* or *non-intersecting*, as shown in Fig. 3. An intersecting spatial relationship ignores objects’ physical information and, as such, the objects can partially intersect with or contain others. In contrast, a non-intersecting relationship disallows the intersections between the objects, which means at best two objects can touch each other (but not intersect). Both spatial relationships are considered in this work. For simplicity, we

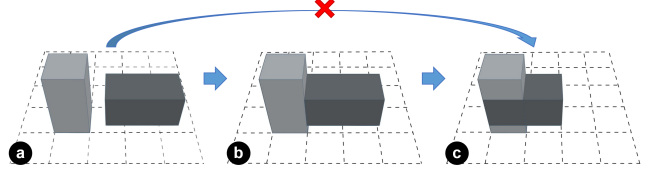


Figure 3: An example of align left: (a) Objects’ initial positions. (b) Non-intersecting alignment. (c) Intersecting alignment. Note that an intersection in a direction can only happen when the objects have been aligned in that direction.

define that an intersection in a direction can only be established if the objects have been aligned in that direction. In other words, an intersection can only happen after alignment.

A general procedure for performing a group-based alignment involves two main phases: (1) *Selection phase*: In this phase, the user first needs to select a group of target objects. (2) *Alignment phase*: the user needs to apply one or more align constraint(s) iteratively to the selected objects until all the requirements are satisfied. In this work, our primary focus is on the second alignment phase.

4 DESIGN CRITERIA

Given the challenge of 3-DOF group-based alignment in VR as defined above, we specified three design criteria (denoted as C#) that such an alignment technique must meet to be effective.

- **C1: Applicability.** The proposed interaction techniques should be applicable within current VR HMDs and immersive systems and should suit commercial VR modeling or content creation applications (e.g., Blocks [16] or SketchBox [17]). Thus, we considered and designed the techniques that are usable with the ubiquitous hand-held controllers, which are represented as virtual hands in the VE.
- **C2: Near-field Interaction.** The proposed technique for 3-DOF alignments should always be within arm’s reach so that a user can easily activate it. A near-field interaction technique can reduce human errors compared to interacting at a distance [19, 40]. Furthermore, it avoids unnecessary body movements, leading to improved efficiency and reduced physical demands.
- **C3: Disambiguation.** *Efficiency*, *learnability*, and *memorability* are three *usability criteria* that any interactive system should meet [28] (pp. 25). A technique with clear operational affordances supports efficient interaction and helps users learn how to use it quickly. We thus made the interaction mechanism consistent in each technique, while allowing the operations towards different alignments to vary naturally. For example, though the gestures in a gesture-based technique may (spatially) vary for different alignments, their semantic interpretations share the same concept.

5 DESIGN OF TECHNIQUES

In this work, we designed and implemented four techniques for group-based 3-DOF alignment in virtual environments: (1) *Align-Panel*, (2) *AlignWidget*, (3) *AlignPin*, and (4) *AlignGestures*. All assume that the user has first selected the group of objects that are to

be aligned. Fig. 4 shows an example of using these techniques to complete an align-right task.

5.1 Align Panel

AlignPanel is a 2D menu-based interface, with icons and texts and uses a layout typical for desktop applications. We created icons with 3D illustrations to represent the various alignments along the x, y, and z-axis, as shown in Fig. 4a. A user can then click on a button to achieve the corresponding alignment. Once selected, a button is highlighted to also indicate the current alignment state in the user interface.

5.2 Align Widget

AlignWidget is a widget-based technique. Overall, its appearance is similar to a typical 3D widget with virtual handles [6, 24]. It consists of six outer handles, three inner axis sticks, and a central cube. For each of the three axes, the two outer handles and the axis stick represent the two ‘side’ alignments and the central alignment, respectively. Tapping on a handle or a stick then aligns the selected objects based on their spatial arrangement relative to the widget. For example, tapping on the right-most handle will align the objects to the right, as shown in Fig. 4b and e. The handles and sticks are colored by axes: red for the x-axis, green for the y-axis, and blue for the z-axis. An activated alignment is indicated by highlighting the borders of the touched handle or stick. To prevent users from false activations of central alignments, we used sticks through the widget center instead of handles, as this approach increases the distances between each widget component. Additionally, we also placed an inactive grey cube in the center. Although *AlignWidget* is movable, our current implementation for the user study cannot be rotated. This ensures that the representations of the alignment will always match the global axes, thereby minimizing confusion.

5.3 Align Pin

Our *AlignPin* technique is an adapted version of a technique from TinkerCAD [15], a 3D modeling web application. When a user opens *AlignPin* after selecting a group of objects, *AlignPin* creates a bounding box for the group and shows three interactive pins along three edges of that bounding box. The user can then touch one of the corresponding pin heads to align the objects. We use 3D pins instead of the 2D ‘flat’ pins that TinkerCAD uses because the 3D ‘style’ makes the pins not only more visible from different viewing directions but also easier to interact with in the VE. The indication for an active alignment is the same as in *AlignWidget*. To meet our design criterion C2, we created a hand-held replica of the *AlignPin* widget. While this looks on the surface similar to a World-In-Miniature approach [34], we only replicate the widget and not the objects to avoid occlusions (see Fig. 4c). Similar to *AlignWidget*, *AlignPin* allows translational movements but not rotational movements.

5.4 Align Gesture

With *AlignGesture*, the user performs a hand gesture to align the selected objects. We identified nine hand gestures that match the alignment requirements, as shown in Fig. 5. We aimed to make the

gestures semantically meaningful and easy to understand. The underlying logic is to move the hand toward the implicitly constructed constraint surface. To perform a side alignment, a user then only needs to keep one hand reasonably still (which then represents the constraint) and move the other hand close to it [26]. On the other hand, the constraint for middle alignments lies between the two hands. Thus, moving two hands towards each other along the target axis specifies an align-middle action¹. In our implementation and to prevent false positives, the user needs to hold the grip buttons on both controllers simultaneously to perform an *AlignGesture*. In addition, the user needs to release the buttons after the gesture, to be able to perform potential other alignment actions (as described in Section 3).

We fine-tuned our gesture recognition through a pilot study with 5 participants, which collected hand movement data. The participants performed each gesture 5 times. Based on the collected data, we were able to identify good thresholds for hand movement distances to recognize the gestures reliably.

6 USER STUDY

The aim of our study was to compare and evaluate the four proposed techniques for 3-DOF group-based alignment in VR environments. We followed the *VR object selection and manipulation study checklist* provided by Bergström et al. [2] to design and report the study.

6.1 Participants

We recruited 20 participants (9 females, 11 males) with ages between 19 to 31 years ($mean = 22.60, SD = 2.50$). All were university students and volunteered to participate in this user study. Based on the results from a questionnaire applied before the experiment, all of them reported that they were right-handed and had normal or correct-to-normal vision. Fifteen participants were familiar or very familiar with VR HMDs, and ten participants were familiar or very familiar with 3D content design or creation applications (such as CAD, 3ds Max, Photoshop, etc.). Fourteen participants used a VR HMD every month or week. Five participants used 3D content design or creation applications on a monthly basis, and seven used them on a weekly or daily basis.

6.2 Apparatus and Experiment Setup

An Oculus Quest 2 standalone VR HMD was used to display the virtual environment (VE). This HMD has a display resolution of 1832×1920 per eye, 89° horizontal field of view, and a 120Hz refresh rate. The HMD was connected to a Windows 10-based desktop with an Intel i7-8700K CPU @ 3.70GHz, an NVIDIA GeForce GTX 1080 Ti GPU, and 16 GB of RAM. Two Oculus controllers were used for interaction with the VE. Our software was developed using Unity (version 2021.3.36f1c1) with the SteamVR Plugin (version 2.7.3). The study was conducted in an empty room. The participants were standing while performing the tasks.

¹Note that the gestures for the alignments along the x-axis are typically unique due to the nature of people’s hands (i.e., left and right hands). In contrast, the gestures for alignment in the y- or z-axis can be performed in two different ways. For example, a user can position their right hand below and move the left hand close to it, or vice versa, to perform an align-bottom action.

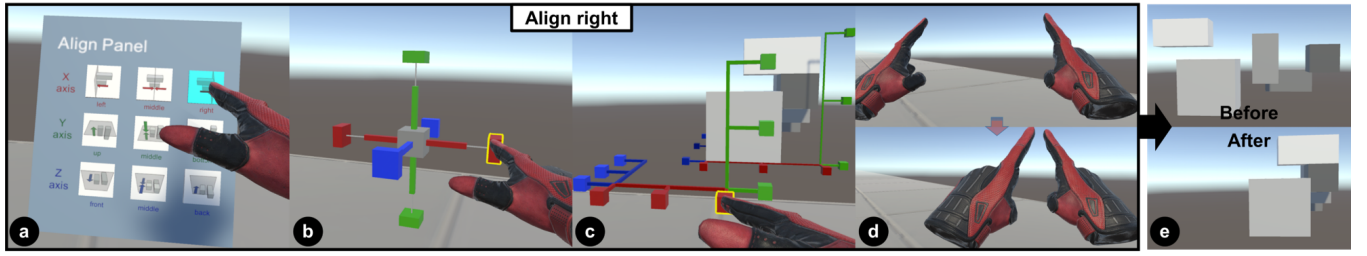


Figure 4: For an align-right task, this figure shows the user interfaces of the four investigated techniques: (a) *AlignPanel*, the user clicks on the button; (b) *AlignWidget*, the user taps on the handle; (c) *AlignPin*, the user taps on the pin head; and (d) *AlignGesture*, the user performs the hand gesture. The resulting alignment is shown in the bottom panel (e). Note that we are showing a non-intersecting alignment, thus the medium-grey front-most bottom block aligns only to the dark grey block to avoid interpenetration.

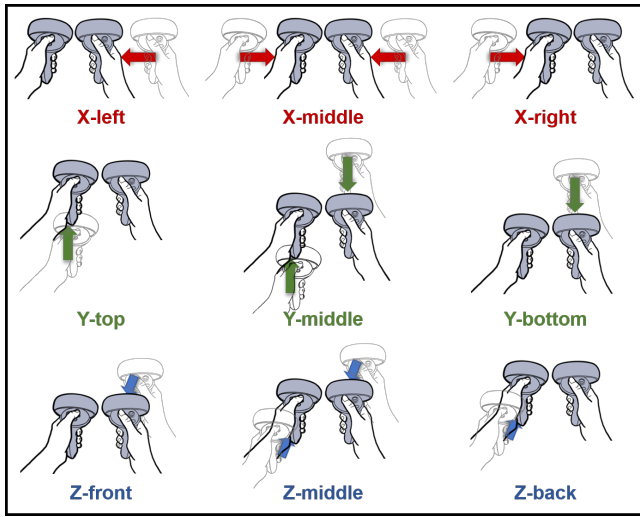


Figure 5: The gesture set for *AlignGesture*, where each of the nine gestures represents one of the potential 3-DOF object alignments in a way that matches the geometric relationships of the alignment operation.

In the experiment, participants saw five objects that they were asked to align and a task specification to specify the alignment requirements (see Fig. 6a and b). To simplify the task, we used cuboids that were within view and within reach from the initial position. To eliminate a potential experimental confound, participants could not directly select a cuboid to translate, rotate, or scale. We used text descriptions in the task specification rather than images or 3D models to reduce the cognitive load for planning and decision-making because such cognitive load could lead to reduced user performance and also negatively affect the experience of using a technique, which could bias the results.

Participants interacted with the VE via the hand-held controllers. To undo an action, users needed to press Button ‘A’ on the right controller, which would move all objects to their previous positions, i.e., to their state before the last action. For *AlignPanel*, *AlignWidget*,

and *AlignPin*, Button ‘Y’ on the left controller toggled the corresponding widget on or off, while Button ‘X’ on the left controller toggled the widget following the participant’s left, non-dominant hand movements, which we identified in our pilot study as a usable method that worked well. By default, the widgets for these three techniques were off and, when turned on, would move with the participant’s left hand. As mentioned previously, for *AlignGesture* a user needed to press and hold the grip buttons of two controllers to perform the corresponding alignment action.

6.3 Design, Task, and Procedure

Our study used a 4×2 within-subjects design with two independent variables: *TECHNIQUE* (*AlignPanel*, *AlignWidget*, *AlignPin*, and *AlignGesture*), and *ALIGNMENTS* (*one-time* and *three-time*). *ALIGNMENTS* varies the number of alignments that participants needed to perform in a task: *one-time* means one requirement (e.g., middle alignment along the z-axis in Fig. 6a), while *three-time* would require three different alignments, one for each axis (e.g., middle alignment along the z-axis, then bottom alignment along the y-axis, and finally middle alignment along the x-axis; see Fig. 6b). The alignment tasks were presented randomly.

The task required participants to align the objects according to the task specification shown above the scene. As mentioned in Section 3, all the cuboids were already selected at the beginning of a trial. A task then ended when the participants had completed the required alignment action(s). The tasks involved both non-intersecting and intersecting alignments. For an intersecting alignment task, the only alignment in the *one-time* task and the third alignment in the *three-time* task involved an intersecting alignment (Fig. 6b).

The order of *TECHNIQUE* conditions was counterbalanced using a Latin-Square design. Following general experimental design recommendations to present tasks in order of complexity, the *one-time* tasks were presented before the *three-time* tasks for each technique. Each *ALIGNMENTS* condition involved 6 alignment tasks, including 3 intersecting alignment tasks and 3 non-intersecting alignment tasks. In total, we collected data from 20 participants $\times$ 4 *TECHNIQUE* $\times$ 2 *ALIGNMENTS* $\times$ 6 alignment tasks = 960 tasks.

Participants were first informed of the purpose of the study and completed a pre-questionnaire collecting their demographic and

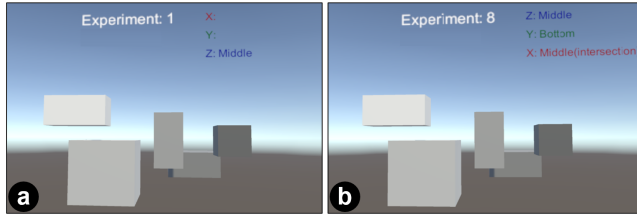


Figure 6: (a) An example of a *one-time* alignment task. (b) An example of a *three-time* alignment task where the third axis (i.e., the third sub-task) is an intersecting alignment task.

past experience information. Then, they were briefed about the VE and the four new interaction techniques, followed by a 10-minute ‘warm-up’ session with the VR device to get accommodated to the VE and the control schemes for each technique. The experiment involved four sessions, corresponding to the four techniques. For each session and to ensure that participants understood the task and current interaction technique well, four training trials, including two *one-time* intersecting and two *three-time* intersecting alignment trials, were given before the formal trials. After each session, we asked participants to record their subjective feelings about the just-used technique through ratings that covered different aspects (introduced in the next section). In addition, the participants were encouraged to take a short break between two sessions. At the end of the experiment, the participants received a short interview. The whole experiment lasted approximately 35 minutes for each participant.

6.4 Evaluation Metrics

We measured the performance of the techniques in terms of *number of failures*, and *completion time*. We defined trials where participants needed to resort to undo actions as failed trials. *Number of failures* is the number of failed trails for each `TECHNIQUE`×`ALIGNMENT` condition. *Completion time* is the time to complete a given alignment task in seconds. We did not consider the failed trials in this measurement because the number of undo actions can confound this time and the reasons for participants performing undos were not always clear. We recorded the time from when the participant first pressed the grip button(s) to perform an *AlignGesture* or pressed the ‘Y’ button for the first time to show the widget for the other three techniques until the required alignment had been achieved.

In addition, we collected participants’ subjective ratings after each session. We used the Raw NASA-TLX [13] to assess the required workload to complete a given task and a positive System Usability Scale (SUS) [20] to evaluate the usability of the techniques. At the end of the experiment, we conducted a semi-structured interview where we asked participants about their experience with the new techniques with respect to different aspects, including (1) the ranking of the techniques based on their overall preference and the reasons for giving this ranking, if they could express them; (2) the strengths and weaknesses of each technique; and (3) any possibilities that they could think of for using or extending the techniques for 6-DOF alignments (i.e., manipulations involving 3-DOF rotations).

6.5 Hypotheses

Based on our literature review, our design process, and pre-experiment pilot trials, we formulated the following hypotheses:

- **H1.** *Participants’ performance in terms of the number of failures and completion time in the given tasks would not vary significantly among the four proposed techniques.* In the design of the techniques, we followed our three design criteria. We hypothesize that all four techniques are usable and support efficient interaction (C1, C2). For each technique, the actions for different alignments are easy to distinguish (C3).
- **H2.** *Participants experience more workload to complete the given tasks with AlignGesture than with the other three techniques.* Participants interact with *AlignPanel*, *AlignWidget*, and *AlignPin* through a virtual tool using simple actions (i.e., tapping a button). Compared to them, *AlignGesture* might experience relatively higher mental effort since participants have to recall and perform the corresponding gesture.
- **H3.** *The perceived usability of AlignPanel is higher than the other three techniques.* Given that *AlignPanel* is adapted from a typical 2D UI, participants might be more familiar with it and thus find it easy to learn and intuitive to use, both of which are important measures of the technique’s usability [20]. Thus, *AlignPanel* could be rated more usable for our well-controlled but reasonably straightforward tasks.

7 RESULTS

The results were analyzed and visualized using R (version 4.2.1) and RStudio (version 2022.02.3).

7.1 Objective Measures

Among the 960 trials we recorded, we considered 104 trials as failed (completed with undo), and these were counted as the number of failures. For the remaining 856 successful trials, we removed 6 trials (0.70%) as outliers due to extreme completion times. For the number of failures and completion time, we performed two-way repeated measures (RM-)ANOVA tests. According to the QQ plots, completion time was normally distributed, while the number of failures was not. Thus, we applied Aligned Rank Transform [7, 41] to the number of failures before the RM-ANOVA tests. If a significant difference was found, pairwise comparisons were conducted with Bonferroni corrections. When the assumption of sphericity was violated, we report the degrees of freedom with Greenhouse-Geisser corrections ($\epsilon < 0.75$ in our cases). We report performance separately for each type of `ALIGNMENT` task, as the interaction between `ALIGNMENT` and `TECHNIQUE` may not provide meaningful insights.

7.1.1 Number of failures. The average number of failures was statistically different for `TECHNIQUE` ($F_{3,133} = 4.515, p = 0.005, \eta_p^2 = 0.092$) and `ALIGNMENT` ($F_{1,133} = 19.624, p < 0.001, \eta_p^2 = 0.129$). There was no significant interaction between `TECHNIQUE` and `ALIGNMENT` in the number of failures ($F_{3,133} = 2.130, p = 0.099, \eta_p^2 = 0.046$). Post-hoc tests did not identify significant differences between `TECHNIQUE` in *one-time* tasks or *three-time* tasks ($p > 0.05$). Fig. 7a shows the results.

7.1.2 Completion time. There was a significant main effect of `ALIGNMENT` on completion time ($F_{1,19} = 291.098, p < 0.001, \eta_p^2 =$

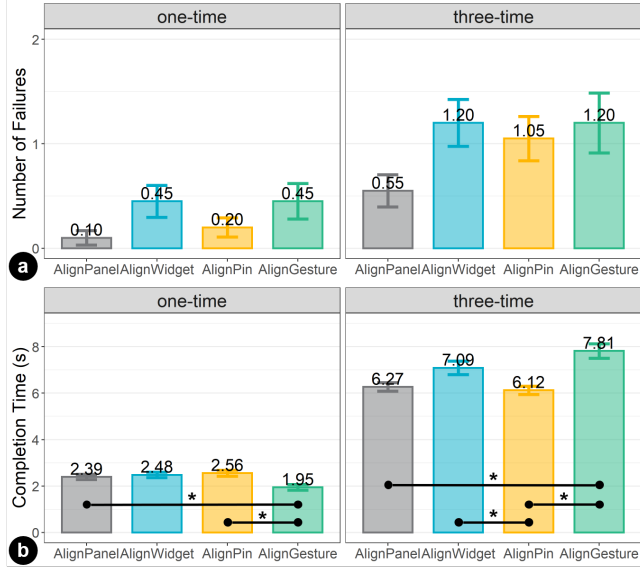


Figure 7: Results of (a) number of failures (failed trials), and (b) completion time of successful trails by TECHNIQUE and ALIGNMENT. Error bars represent standard errors. Statistical significant effects are marked with *, **, and *, representing a significance level of 0.05, 0.01, and 0.001, respectively. (The same marking scheme is used in the other figures, too.)**

0.939), with a significant interaction between TECHNIQUE and ALIGNMENT ($F_{2,38.08} = 12.176, p < 0.001, \eta_p^2 = 0.391$). However, completion time was not significantly different between TECHNIQUES ($F_{1.74,33.11} = 2.344, p = 0.118, \eta_p^2 = 0.110$). As mentioned, we analyzed the data for *one-time* and *three-time* tasks separately to dig deeper. The pairwise tests identified that the main effect of TECHNIQUE was significant for both *one-time* and *three-time* alignment tasks ($p = 0.014$ and $p = 0.007$, respectively). Pairwise comparisons showed that for the *one-time* task, the completion time of using *AlignGesture* was significantly shorter than *AlignPanel* ($p = 0.018$) and *AlignPin* ($p = 0.045$). For the *three-time* task, *AlignPin* was significantly faster than *AlignWidget* ($p = 0.038$) and *AlignGesture* ($p = 0.019$). *AlignPanel* is significantly faster than *AlignGesture* ($p = 0.026$) as well. The results are summarized in Fig. 7b. The average completion time for a *three-time* task is 2.62 times of the completion time for a *one-time* task using *AlignPanel*, 3.22 times using *AlignWidget*, 2.39 times using *AlignPin*, and 4.01 times using *AlignGesture*.

7.2 Subjective Measures

The sub-scales of the NASA-TLX were scaled to a range from 0 to 100, the same as their weighted overall scores. We applied Aligned Rank Transform [7, 41] for non-parametric factorial analyses for the NASA-TLX scores and SUS scores.

7.2.1 NASA-TLX. There was a significant effect of TECHNIQUE on overall workload scores ($F_{3,57} = 6.778, p < 0.001, \eta_p^2 = 0.162$). For the sub-scales of the NASA-TLX, we found significant effects for mental demand ($F_{3,57} = 7.969, p < 0.001, \eta_p^2 = 0.167$), physical

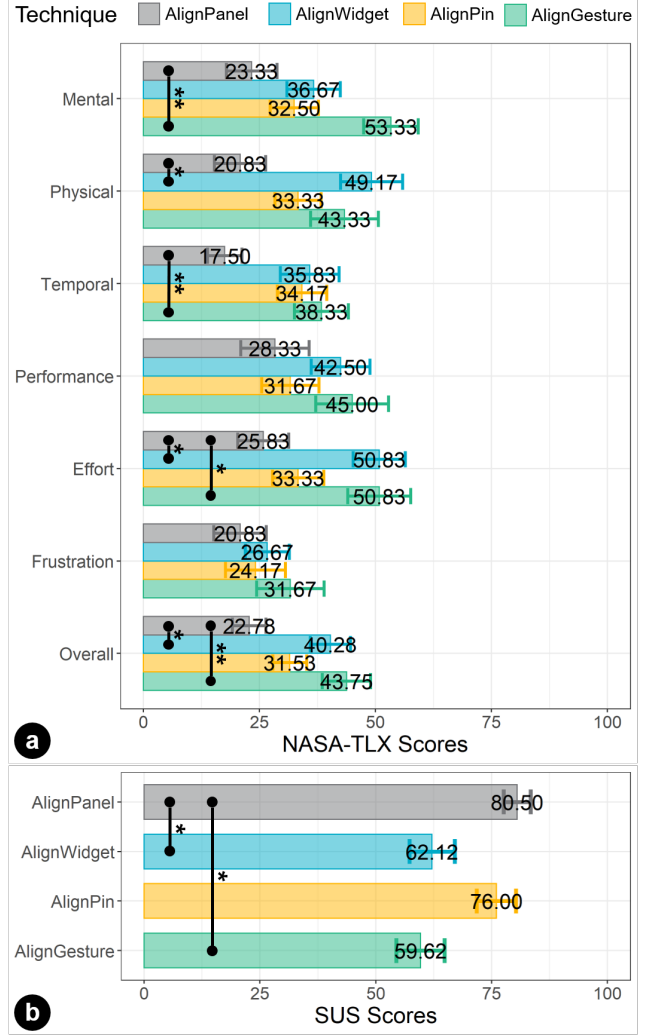


Figure 8: (a) NASA-TLX scores and (b) SUS scores by TECHNIQUE.

demand ($F_{3,57} = 6.627, p < 0.001, \eta_p^2 = 0.137$), temporal demand ($F_{3,57} = 4.799, p = 0.005, \eta_p^2 = 0.109$), and effort ($F_{3,57} = 5.283, p = 0.003, \eta_p^2 = 0.155$). Fig. 8a shows the results and any significant differences from pairwise comparisons.

7.2.2 SUS. There was a significant effect of TECHNIQUE on overall SUS scores ($F_{3,57} = 7.488, p < 0.001, \eta_p^2 = 0.176$). Pairwise comparisons revealed that the perceived usability of *AlignPanel* was significantly higher than *AlignWidget* ($p = 0.025$) and *AlignGesture* ($p = 0.008$). Fig. 8b shows the results.

7.2.3 Rankings. Fig. 9 summarizes the subjective rankings in terms of overall preference. We present and discuss all other interview responses in the Discussion section.

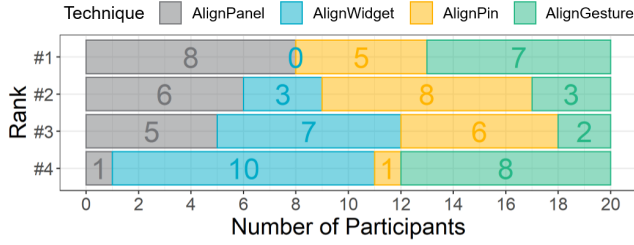


Figure 9: Rankings of the four proposed techniques in terms of overall preference

8 DISCUSSION

8.1 Technique Evaluation

Overall, hypotheses **H1** and **H3** were partially supported by our results. We found the number of failures did not vary significantly, but the completion time did (**H1**). *AlignPanel* was perceived more usable than *AlignWidget* and *AlignGesture*, but not *AlignPin* (**H3**). **H2** was not supported, as *AlignGesture*'s overall workload score was only significantly higher than *AlignPanel*. Based on these results, we discuss the techniques separately in this section.

8.1.1 AlignPanel. *AlignPanel* is a technique inspired by and adapted from desktop applications. Based on participants' comments, it was "intuitive" and "easy to learn" ($N = 7$). Compared to the other proposed techniques, *AlignPanel* induced only a relatively low workload to complete a 3-DOF group-based alignment task, regardless of the task requirements, as can be seen from Fig. 8a. In addition, it received the highest overall SUS score, significantly higher than *AlignWidget* and *AlignGesture*, which partially supports our **H3** and indicates high usability compared to the other techniques. Yet, in a *one-time* task, the completion time of using *AlignPanel* was significantly longer than using *AlignGesture* but we believe this difference does not affect *AlignPanel*'s practical use. Based on these benefits, eight participants (40%) rated *AlignPanel* as the most preferred technique. Only one participant (P6) ranked it last because this participant felt the experience of interacting with *AlignPanel* to be dull, and four other participants reported a similar sentiment. In addition, P8 mentioned: "I felt like I was not working in a VR or even a 3D environment when using *AlignPanel*." P19 also mentioned that the interactivity of *AlignPanel* was low.

8.1.2 AlignWidget. Overall, the participants did not like *AlignWidget*, and half of them ranked it the lowest. In terms of completion time and the number of failures, there was no significant difference between *AlignWidget* and the other techniques. However, thirteen participants felt *AlignWidget* was "hard to use", and four of them complained about the extra caution needed for interacting with the middle or back widget for the z-axis, i.e., the part of the *AlignWidget* that points away from the viewer. This might be the reason that the workload scores of this technique in terms of physical demand and effort were, on average, the highest among the four techniques. Similarly, the overall usability of *AlignWidget* was also perceived as low compared to the other techniques. Although some participants ($N = 4$) suggested a slight rotation of the widget (relative to the global axes) might make the farthest parts of the widgets easier

to interact with, they also shared the concern that such rotation may generate confusion. In terms of the positive aspects of this technique, two participants (P1, P3) mentioned *AlignWidget* as a good presentation of the 3D relationships.

8.1.3 AlignPin. Interestingly, *AlignPin* was slow in *one-time* tasks but fast in *three-time* tasks (see Fig. 7b). One possible reason is that participants needed to first locate the correct pin head among the separated axes, which is relatively time-consuming compared to the other techniques in an *one-time* task; while since it was "easy to understand" ($N = 4$) with "representative visuals" ($N = 5$), participants were able to complete multiple alignments without a complicated thought process, which yielded a shorter completion time in the *three-time* tasks. On the whole, *AlignPin* was highly usable, required a low workload, and was preferred by the participants for doing a group-based alignment task.

8.1.4 AlignGesture. In contrast to *AlignPin*, *AlignGesture* was fast in the *one-time* tasks but slow in the *three-time* tasks. Unlike the previous three techniques, *AlignGesture* did not require participants to locate and select a target alignment option, so it could be performed quickly when users accurately recalled the correct gesture. Though the overall workload of *AlignGesture* was only significantly lower than *AlignPanel* (which did not support **H2**), we found that it was harder to learn and to remember how to use it, compared to the other techniques. This led to high mental demand and effort, as well as low usability (see Fig. 8). Especially when doing an iterative *three-time* task, the cognitive load was relatively high for recalling the gesture(s). Nevertheless, we found seven participants ranked it as the most preferred technique and three ranked it as the second-most preferred, which together is half the size of our participant pool. Participants gave this high ranking mainly because they believed if they used it for a longer time, *AlignGesture* would be the fastest and the most natural technique ($N = 7$).

8.2 Design Choices and Recommendations

Current VR modeling applications, such as Sketchbox [17] and Blocks [16], use an in-hand menu-based interface, which makes them easy to integrate *AlignPanel*. However, when asked about its use for 6-DOF alignment tasks, twelve participants said it was not suitable or could not be used. Indeed, using a 2D menu to represent and control complex 3D relationships in immersive environments might be less intuitive. Thus, ***AlignPanel* is only appropriate when the use case requires 3-DOF translational alignments and when the users' view matches or roughly matches the global axes.**

For the 3D widget-based techniques, all participants agreed that both *AlignWidget* and *AlignPin* could be used for 6-DOF alignment tasks as long as the rotation of the widget matches the rotation of the object group. In addition, it is also possible to use these widgets directly to control the rotation of the object group, simply by rotating the widget. Based on our results, **we suggest *AlignPin* to be used in practical applications**, while *AlignWidget* is not recommended due to its relatively low usability. Compared to *AlignPanel*, *AlignPin* could have broader uses in immersive VR environments.

Based on the collected feedback, *AlignGesture* is a 'controversial' technique. There is no denying that *AlignGesture* has a high learning

cost, even if a subset of users may feel that such gestural input is natural and intuitive. As for using it for 6-DOF alignment tasks, participants felt it was easier to use for such scenarios because they could perform the gesture along the rotated axes. However, three participants (P4, P10, P16) raised concerns about the precision of hand movement recognition. Given the above aspects, **we do not recommend using *AlignGesture* or only integrating it as an additional option for long-term users who prefer to use it.**

Though most participants chose not to toggle the current alignment widget following their hand movement, we still recommend giving this choice to users in real application scenarios. Our experimental trials involved two types of alignment tasks, with and without intersections. We did not observe a difference between these types during the experiment or from the collected results, which was in line with our expectations as repeated actions were necessary for these conditions. Thus, it is possible to use the same mechanism in real scenarios. Another alternative is to make the type of intersection a system option so that users can choose to activate or deactivate the intersection behavior for the following sets of manipulations.

This work explored techniques for 3-DOF group-based object alignment where the alignment constraints are implicitly formed based on the object group. By adding a step for specifying alignment constraint(s) before the alignment phase (see Section 3), we believe that our proposed techniques are still applicable to other types of alignments as long as all the selected objects need to match the same alignment goal.

Though our user study was conducted using a VR HMD, we expect that our results and the derived design choices and recommendations could inspire and be beneficial to the design and development of applications in other immersive systems, such as CAVE or AR HMDs. Our results could also generalize to other types of applications in immersive systems which involve similar group manipulations. For example, VR world-building games can benefit from our research, as our techniques enable players to precisely align large numbers of structural elements, which supports the quick creation and modification of larger structures. Another example could be collaborative productivity applications, where users might want to align cluttered notes, to-do items, mind map blocks, or other visual elements from different collaborators, to make the hierarchy or flow better visible. We thus believe that our techniques allow efficient and accurate group-based alignments of visual elements across many VR applications.

8.3 Limitations and Future Work

We identified the following limitations, which could point to directions for future work. First, we used a fairly simple strategy for gesture recognition, which also yielded reasonably robust recognition. Although we did not observe any recognition errors during the user study, other approaches such as Dynamic Time Warping [3] could further improve the precision and enable its use in a larger variety of contexts. Second, as the first exploration of group-based object alignment in VR, we used a controlled experimental design and limited the task to the alignment phase with translational movements. Future work could evaluate the techniques by giving participants more freedom to play with the techniques and also to

test performance with more complex alignment tasks, including 6-DOF ones, to see if the techniques provide a smooth workflow for such scenarios too. Furthermore, it is worth investigating whether or how our techniques could coexist with other VR selection and manipulation techniques that are optimized for performance and complex scenarios [44–46], or allow for rapid mode-switching in interaction [31, 36]. Third, we would like to enhance and embed our techniques into existing VR modeling applications [16, 17] to compare their performance with a baseline approach in real use cases, which at the moment typically involves the manipulation of only a single object at a time. Finally, most of our participants were familiar with VR headsets. A more varied participant pool could provide additional insights, especially when the tasks and/or the techniques become more challenging, as mentioned above.

9 CONCLUSION

In this work, we explored group-based alignment in immersive virtual reality (VR) systems, primarily focusing on three degrees of freedom (3-DOF) translations. We first identified three design criteria that a technique must meet to address the alignment of a group of objects in VR, and then proposed four interaction techniques: *AlignPanel*, *AlignWidget*, *AlignPin*, and *AlignGesture*. They were evaluated via a user study with 20 participants in terms of their performance, workload, and usability. Our results showed that using *AlignPanel* and *AlignPin* induced less workload and were considered highly usable. *AlignWidget* received negative feedback and thus is not suggested for practical use. Though participants generally rated *AlignGesture* worse in workload and usability, they also saw some benefits in this technique. Based on these findings, we provided several design recommendations for using the four alignment techniques in VR applications. Our alignment techniques and the recommendations derived from the user study can help researchers and designers develop future VR object alignment techniques that can be integrated into a variety of application scenarios.

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REFERENCES

- [1] Mayra D. Barrera-Machuca, Alvaro Cassinelli, and Christian Sandor. 2020. Context-Based 3D Grids for Augmented Reality User Interfaces. In *Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20 Adjunct). Association for Computing Machinery, New York, NY, USA, 73–76. <https://doi.org/10.1145/3379350.3416190>
- [2] Joanna Bergström, Tor-Salve Dalsgaard, Jason Alexander, and Kasper Hornbæk. 2021. How to Evaluate Object Selection and Manipulation in VR? Guidelines from 20 Years of Studies. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 533, 20 pages. <https://doi.org/10.1145/3411764.3445193>
- [3] Donald J. Berndt and James Clifford. 1994. Using Dynamic Time Warping to Find Patterns in Time Series. In *KDD Workshop*.
- [4] Eric A. Bier and Maureen C. Stone. 1986. Snap-Dragging. *SIGGRAPH Comput. Graph.* 20, 4 (aug 1986), 233–240. <https://doi.org/10.1145/15886.15912>
- [5] Marianela Ciolfi Felice, Nolwenn Maudet, Wendy E. Mackay, and Michel Beaudouin-Lafon. 2016. Beyond Snapping: Persistent, Tweakable Alignment

- and Distribution with StickyLines. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (Tokyo, Japan) (UIST '16). Association for Computing Machinery, New York, NY, USA, 133–144. <https://doi.org/10.1145/2984511.2984577>
- [6] Brookshire D. Conner, Scott S. Snibbe, Kenneth P. Herndon, Daniel C. Robbins, Robert C. Zeleznik, and Andries van Dam. 1992. Three-Dimensional Widgets. In *Proceedings of the 1992 Symposium on Interactive 3D Graphics* (Cambridge, Massachusetts, USA) (I3D '92). Association for Computing Machinery, New York, NY, USA, 183–188. <https://doi.org/10.1145/147156.147199>
 - [7] Lisa A. Elkin, Matthew Kay, James J. Higgins, and Jacob O. Wobbrock. 2021. An Aligned Rank Transform Procedure for Multifactor Contrast Tests. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 754–768. <https://doi.org/10.1145/3472749.3474784>
 - [8] S. Frees and G.D. Kessler. 2005. Precise and rapid interaction through scaled manipulation in immersive virtual environments. In *IEEE Proceedings. VR 2005. Virtual Reality, 2005*. IEEE, Washington, DC, USA, 99–106. <https://doi.org/10.1109/VR.2005.1492759>
 - [9] Mathias Frisch, Sebastian Kleinau, Ricardo Langner, and Raimund Dachselt. 2011. Grids & Guides: Multi-Touch Layout and Alignment Tools. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 1615–1618. <https://doi.org/10.1145/1978942.1979177>
 - [10] Mathias Frisch, Ricardo Langner, and Raimund Dachselt. 2011. Neat: A Set of Flexible Tools and Gestures for Layout Tasks on Interactive Displays. In *Proceedings of the ACM International Conference on Interactive Tabletops and Surfaces* (Kobe, Japan) (ITS '11). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/2076354.2076356>
 - [11] Michael Gleicher. 1992. Briar: A Constraint-Based Drawing Program. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Monterey, California, USA) (CHI '92). Association for Computing Machinery, New York, NY, USA, 661–662. <https://doi.org/10.1145/142750.143074>
 - [12] Mark Hancock, Sheelagh Carpendale, and Andy Cockburn. 2007. Shallow-Depth 3d Interaction: Design and Evaluation of One-, Two- and Three-Touch Techniques. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '07). Association for Computing Machinery, New York, NY, USA, 1147–1156. <https://doi.org/10.1145/1240624.1240798>
 - [13] Sandra G. Hart. 2006. NASA-Task Load Index (NASA-TLX); 20 Years Later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 50, 9 (2006), 904–908. <https://doi.org/10.1177/154193120605000909>
 - [14] Devamardeep Hayatpur, Seongkook Heo, Haijun Xia, Wolfgang Stuerzlinger, and Daniel Wigdor. 2019. Plane, Ray, and Point: Enabling Precise Spatial Manipulations with Shape Constraints. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 1185–1195. <https://doi.org/10.1145/3332165.3347916>
 - [15] Autodesk Inc. [n.d.]. TinkerCAD. <https://www.tinkercad.com/>. Accessed: 2022-08-04.
 - [16] Google Inc. [n.d.]. Blocks. <https://arvr.google.com/blocks/>. Accessed: 2022-08-04.
 - [17] Sketchbox Inc. [n.d.]. Sketchbox. <https://design.sketchbox3d.com/>. Accessed: 2022-08-04.
 - [18] Joseph J LaViola Jr, Ernst Kruijff, Ryan P McMahan, Doug Bowman, and Ivan P Poupyrev. 2017. *3D user interfaces: theory and practice*. Addison-Wesley Professional, Boston, MA, USA.
 - [19] Chia-Yang Lee, Wei-An Hsieh, David Brickler, Sabarish V. Babu, and Jung-Hong Chuang. 2021. Design and Empirical Evaluation of a Novel Near-Field Interaction Metaphor on Distant Object Manipulation in VR. In *Symposium on Spatial User Interaction* (Virtual Event, USA) (SUI '21). Association for Computing Machinery, New York, NY, USA, Article 13, 11 pages. <https://doi.org/10.1145/3485279.3485296>
 - [20] James R. Lewis. 2018. The System Usability Scale: Past, Present, and Future. *International Journal of Human-Computer Interaction* 34, 7 (2018), 577–590. <https://doi.org/10.1080/10447318.2018.1455307>
 - [21] John Lucas, Doug Bowman, Jian Chen, and Chad Wingrave. 2005. Design and Evaluation of 3D Multiple Object Selection Techniques. (01 2005).
 - [22] Anthony Martinet, G ry Casiez, and Laurent Grisoni. 2010. The Effect of DOF Separation in 3D Manipulation Tasks with Multi-Touch Displays. In *Proceedings of the 17th ACM Symposium on Virtual Reality Software and Technology* (Hong Kong) (VRST '10). Association for Computing Machinery, New York, NY, USA, 111–118. <https://doi.org/10.1145/1889863.1889888>
 - [23] Daniel Mendes, Fabio Marco Caputo, Andrea Giachetti, Alfredo Ferreira, and Joaquim Jorge. 2019. A Survey on 3D Virtual Object Manipulation: From the Desktop to Immersive Virtual Environments. *Computer Graphics Forum* 38, 1 (2019), 21–45. <https://doi.org/10.1111/cgf.13390>
 - [24] Daniel Mendes, Filipe Relvas, Alfredo Ferreira, and Joaquim Jorge. 2016. The Benefits of DOF Separation in Mid-Air 3D Object Manipulation (VRST '16). Association for Computing Machinery, New York, NY, USA, 261–268. <https://doi.org/10.1145/2993369.2993396>
 - [25] Daniel Mendes, Mauricio Sousa, Rodrigo Lorena, Alfredo Ferreira, and Joaquim Jorge. 2017. Using Custom Transformation Axes for Mid-Air Manipulation of 3D Virtual Objects. In *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology* (Gothenburg, Sweden) (VRST '17). Association for Computing Machinery, New York, NY, USA, Article 27, 8 pages. <https://doi.org/10.1145/3139131.3139157>
 - [26] Vijayakumar Nanjappan, Hai-Ning Liang, Feiyu Lu, Konstantinos Papangelis, Yong Yue, and Ka Lok Man. 2018. User-elicited dual-hand interactions for manipulating 3D objects in virtual reality environments. *Human-centric Computing and Information Sciences* 8, 1 (10 2018), 16 pages. <https://doi.org/10.1186/s13673-018-0154-5>
 - [27] Ivan Poupyrev, Mark Billinghurst, Suzanne Weghorst, and Tadao Ichikawa. 1996. The Go-Go Interaction Technique: Non-Linear Mapping for Direct Manipulation in VR. In *Proceedings of the 9th Annual ACM Symposium on User Interface Software and Technology* (Seattle, Washington, USA) (UIST '96). Association for Computing Machinery, New York, NY, USA, 79–80. <https://doi.org/10.1145/237091.237102>
 - [28] Jennifer Preece, Yvonne Rogers, and Helen Sharp. 2019. *Interaction design: beyond human-computer interaction*. Wiley, Hoboken, NJ, USA. 656 pages.
 - [29] Roope Raisamo and Kari-Jouko R ih . 1996. A New Direct Manipulation Technique for Aligning Objects in Drawing Programs. In *Proceedings of the 9th Annual ACM Symposium on User Interface Software and Technology* (Seattle, Washington, USA) (UIST '96). Association for Computing Machinery, New York, NY, USA, 157–164. <https://doi.org/10.1145/237091.237113>
 - [30] Warren Robinett and Richard Holloway. 1992. Implementation of Flying, Scaling and Grabbing in Virtual Worlds. In *Proceedings of the 1992 Symposium on Interactive 3D Graphics* (Cambridge, Massachusetts, USA) (I3D '92). Association for Computing Machinery, New York, NY, USA, 189–192. <https://doi.org/10.1145/147156.147201>
 - [31] Rongkai Shi, Nan Zhu, Hai-Ning Liang, and Shengdong Zhao. 2021. Exploring Head-based Mode-Switching in Virtual Reality. In *2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 118–127. <https://doi.org/10.1109/ISMAR52148.2021.00026>
 - [32] Graham Smith, Tim Salzman, and Wolfgang Stuerzlinger. 2001. 3D Scene Manipulation with 2D Devices and Constraints. In *Proceedings of Graphics Interface 2001* (Ottawa, Ontario, Canada) (GI '01). Canadian Information Processing Society, CAN, 135–142.
 - [33] Peng Song, Wooi Boon Goh, William Hutama, Chi-Wing Fu, and Xiaopei Liu. 2012. A Handle Bar Metaphor for Virtual Object Manipulation with Mid-Air Interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 1297–1306. <https://doi.org/10.1145/2207676.2208585>
 - [34] Richard Stoakley, Matthew J. Conway, and Randy Pausch. 1995. Virtual Reality on a WIM: Interactive Worlds in Miniature. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '95). ACM Press/Addison-Wesley Publishing Co., USA, 265–272. <https://doi.org/10.1145/223904.223938>
 - [35] W. Stuerzlinger and G. Smith. 2002. Efficient manipulation of object groups in virtual environments. In *Proceedings IEEE Virtual Reality 2002*. 251–258. <https://doi.org/10.1109/VR.2002.996529>
 - [36] Hemant Bhaskar Surale, Fabrice Matulic, and Daniel Vogel. 2019. Experimental Analysis of Barehand Mid-Air Mode-Switching Techniques in Virtual Reality. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300426>
 - [37] Manuel Veit, Antonio Capobianco, and Dominique Bechmann. 2009. Influence of Degrees of Freedom's Manipulation on Performances during Orientation Tasks in Virtual Reality Environments. In *Proceedings of the 16th ACM Symposium on Virtual Reality Software and Technology* (Kyoto, Japan) (VRST '09). Association for Computing Machinery, New York, NY, USA, 51–58. <https://doi.org/10.1145/1643928.1643942>
 - [38] Yanqing Wang, Christine L. MacKenzie, Valerie A. Summers, and Kellogg S. Booth. 1998. The Structure of Object Transportation and Orientation in Human-Computer Interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Los Angeles, California, USA) (CHI '98). ACM Press/Addison-Wesley Publishing Co., USA, 312–319. <https://doi.org/10.1145/274644.274688>
 - [39] Daniel Wigdor, Hrvoje Benko, John Pella, Jarrod Lombardo, and Sarah Williams. 2011. Rock & Rails: Extending Multi-Touch Interactions with Shape Gestures to Enable Precise Spatial Manipulations. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 1581–1590. <https://doi.org/10.1145/1978942.1979173>
 - [40] Curtis Wilkes and Doug A. Bowman. 2008. Advantages of Velocity-Based Scaling for Distant 3D Manipulation. In *Proceedings of the 2008 ACM Symposium on Virtual Reality Software and Technology* (Bordeaux, France) (VRST '08). Association for Computing Machinery, New York, NY, USA, 23–29. <https://doi.org/10.1145/1450579.1450585>

- [41] Jacob O. Wobbrock, Leah Findlater, Darren Gergle, and James J. Higgins. 2011. The Aligned Rank Transform for Nonparametric Factorial Analyses Using Only Anova Procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (*CHI '11*). Association for Computing Machinery, New York, NY, USA, 143–146. <https://doi.org/10.1145/1978942.1978963>
- [42] Josef Wolfartsberger, Jan Zenisek, Christoph Sievi, and Mathias Silmbroth. 2017. A virtual reality supported 3D environment for engineering design review. In *2017 23rd International Conference on Virtual System & Multimedia (VSMM)*. 1–8. <https://doi.org/10.1109/VSMM.2017.8346288>
- [43] Pengfei Xu, Hongbo Fu, Chiew-Lan Tai, and Takeo Igarashi. 2015. GACA: Group-Aware Command-Based Arrangement of Graphic Elements. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (*CHI '15*). Association for Computing Machinery, New York, NY, USA, 2787–2795. <https://doi.org/10.1145/2702123.2702198>
- [44] Difeng Yu, Hai-Ning Liang, Kaixuan Fan, Heng Zhang, Charles Fleming, and Konstantinos Papangelis. 2020. Design and Evaluation of Visualization Techniques of Off-Screen and Occluded Targets in Virtual Reality Environments. *IEEE Transactions on Visualization and Computer Graphics* 26, 9 (2020), 2762–2774. <https://doi.org/10.1109/TVCG.2019.2905580>
- [45] Difeng Yu, Hai-Ning Liang, Xueshi Lu, Kaixuan Fan, and Barrett Ens. 2019. Modeling Endpoint Distribution of Pointing Selection Tasks in Virtual Reality Environments. *ACM Transactions on Graphics* 38, 6, Article 218 (nov 2019), 13 pages. <https://doi.org/10.1145/3355089.3356544>
- [46] Difeng Yu, Xueshi Lu, Rongkai Shi, Hai-Ning Liang, Tilman Dingler, Eduardo Velloso, and Jorge Goncalves. 2021. Gaze-Supported 3D Object Manipulation in Virtual Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 734, 13 pages. <https://doi.org/10.1145/3411764.3445343>